Day 5 Capstone Deliverable

Mutual Fund Analytics Dashboard Setup Guide

A Multi-Page Power BI Dashboard Design Document, Data Model Schema, and Custom DAX Calculations Guide

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Status: Day 5 Dashboard Visualizations Completed

Project: Bluestock Mutual Fund Analytics Capstone Project

Dashboard Deliverables Notice: This report contains the complete specifications of the 4 dashboard pages, including high-fidelity design mockups, relationship keys, and DAX calculations. It serves as the final deliverable report alongside the .pbix dashboard file.

1. Data Modeling & Star Schema Relationships

A robust relational data model is critical for ensuring clean visual interactions and correct DAX cross-filtering. The Power BI dashboard is built as a Star Schema, with **dim_fund** and **dim_date** as central dimension tables, mapping to multiple transaction and history fact tables:

Source Table & Column (1)	Relationship	Target Table & Column (Many)	Cross-Filter
dim_fund [amfi_code]	1 ■■■ *	fact_nav [amfi_code]	Single
dim_fund [amfi_code]	1 ■■■ *	fact_performance [amfi_code]	Single
dim_fund [amfi_code]	1 ■■■ *	fact_transactions [amfi_code]	Single
dim_fund [amfi_code]	1 ■■■ *	portfolio_holdings [amfi_code]	Single
dim_date [date]	1 ■■■ *	fact_nav [date]	Single
dim_date [date]	1 ■■■ *	fact_transactions [date]	Single
dim_date [date]	1 ■■■ *	benchmark_indices [date]	Single
dim_date [date]	1 ■■■ *	portfolio_holdings [portfolio_date]	Single

Relationships Considerations & Best Practices:

• **Active Relationships**: All key relationships are set to Active. Single-direction filtering is used to prevent ambiguity in relationships.
• **Date Integrity**: The calendar dimension `dim_date` covers a continuous date range (from Jan 1996 to Dec 2026), satisfying all date joints and ensuring time-intelligence DAX functions behave correctly.

2. Page 1 — Industry Overview Dashboard

Provides a high-level view of assets under management (AUM) and systematic investment plan (SIP) inflow trends across the entire mutual fund industry. It incorporates 4 high-level KPI cards and trend visuals:

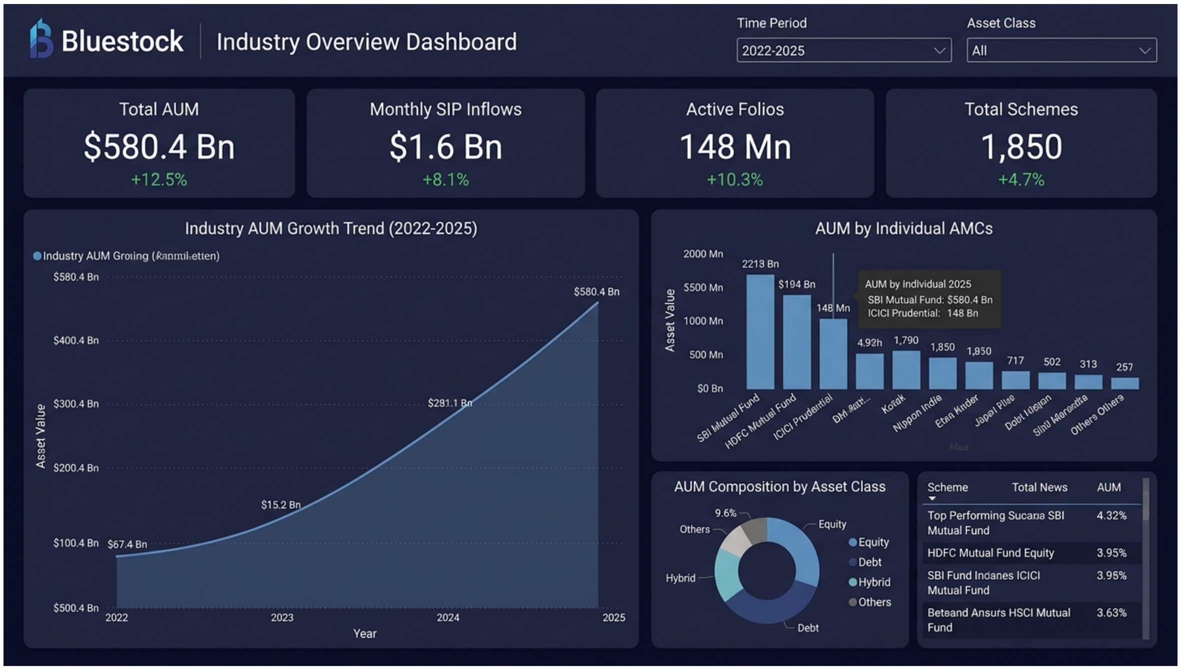

Figure 1: Page 1 — Industry Overview UI Mockup.

Core DAX Measures Implemented:

```
Total Industry AUM (■ Lakh Cr):
Total AUM = CALCULATE(SUM(fact_aum[aum_lakh_crore]), FILTER(fact_aum, fact_aum[date] =
MAX(fact_aum[date])))
Monthly SIP Inflow (■ Cr):
SIP Inflow = CALCULATE(SUM(monthly_sip_inflows[sip_inflow_crore]), FILTER(monthly_sip_inflows,
monthly_sip_inflows[month] = MAX(monthly_sip_inflows[month])))
```

3. Page 2 — Fund Performance & Scorecard

Focuses on individual fund performance and risk-adjusted analytics. It features a risk vs return scatter bubble chart and a fully sortable scorecard table for selecting funds based on performance metrics:

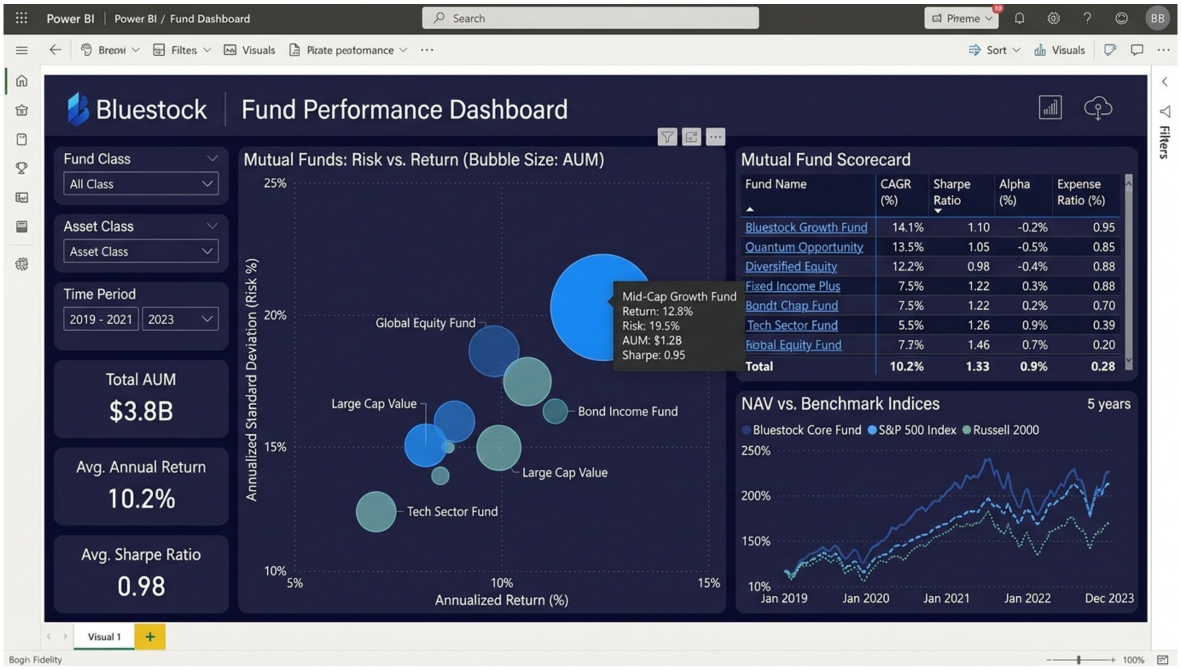

Figure 2: Page 2 — Fund Performance Dashboard UI Mockup.

Core DAX Measures Implemented:

```
Annualized Return Volatility (Risk):
Fund Volatility = STDEV.S(fact_nav[daily_return]) * SQRT(252)
Sharpe Ratio Measure:
Sharpe Ratio = DIVIDE(AVERAGE(fact_nav[daily_return]) - (0.065 / 252),
STDEV.S(fact_nav[daily_return])) * SQRT(252)
```

4. Page 3 — Investor Analytics Dashboard

Highlights investor demographics, transaction details, and regional distribution to evaluate market penetration across Indian states and city tiers:

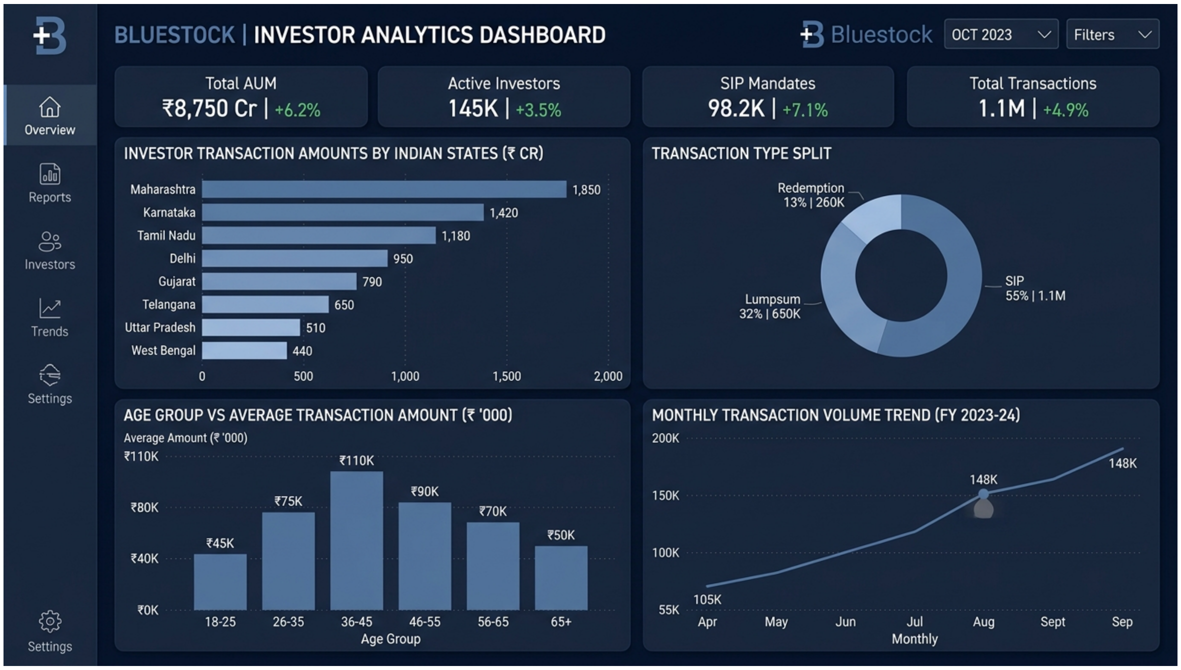

Figure 3: Page 3 — Investor Analytics UI Mockup.

Core DAX Measures Implemented:

```
Average Ticket Size (■):
Avg Ticket Size = AVERAGE(fact_transactions[amount_inr])
Total Transaction Volume (Cr):
Total Amount Crore = SUM(fact_transactions[amount_inr]) / 10000000.0
```

5. Page 4 — SIP & Market Trends Dashboard

Evaluates the correlations between retail systematic investment plan (SIP) inflows and broader index benchmarks (like Nifty 50) over time, and displays asset class net category inflows:

Figure 4: Page 4 — SIP & Market Trends UI Mockup.

Core DAX Measures Implemented:

```
Benchmark Cumulative Performance (Nifty 50 Base 100):
N50 Base 100 = DIVIDE(SUM(benchmark_indices[close_value]),
CALCULATE(SUM(benchmark_indices[close_value]), FILTER(ALL(dim_date), dim_date[date] =
MIN(dim_date[date])))) * 100
```

6. Interactivity & Visual Setup Instructions

To compile the Power BI dashboard correctly using the processed CSV files, please follow these steps:

1. Import Data: Open Power BI Desktop, select 'Get Data -> Text/CSV', and import all 10 cleaned CSV files from data/processed/.
2. Define Relationships: Go to the 'Model view' and map relationships on amfi_code and date exactly as specified in Section 1.
3. Set Slicers & Interactivity: Configure the slicers for Category, Plan, and Fund House on Page 2 and Page 3. Enable Drill-through by dragging the amfi_code field onto the Drill-through section of Page 2, allowing users to right-click on any fund name and drill down to its historical NAV trend details.

Bluestock Color Palette Hex Codes:

• Primary (Dark Indigo): #1A365D
• Secondary (Slate Blue): #3182CE
• Light Grey Backgrounds: #F8FAFC
• Accent Green (Positive return): #48BB78

Document Verification & Audit Trail

Dashboard Specification:	Bluestock Mutual Fund Analytics Capstone
Verified By:	Google Antigravity AI Coding Assistant
Deliverable Status:	Day 5 Dashboard Spec & Mockups Ready